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Department of CSE

B.Tech. CSE Trimester -5

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Cloud Infrastructure and Services
25CS2206E

LAB RECORD

2026

Experiment -1 (b)

AIM: Enabling Multi-Factor Authentication to Secure Your Access

Description:

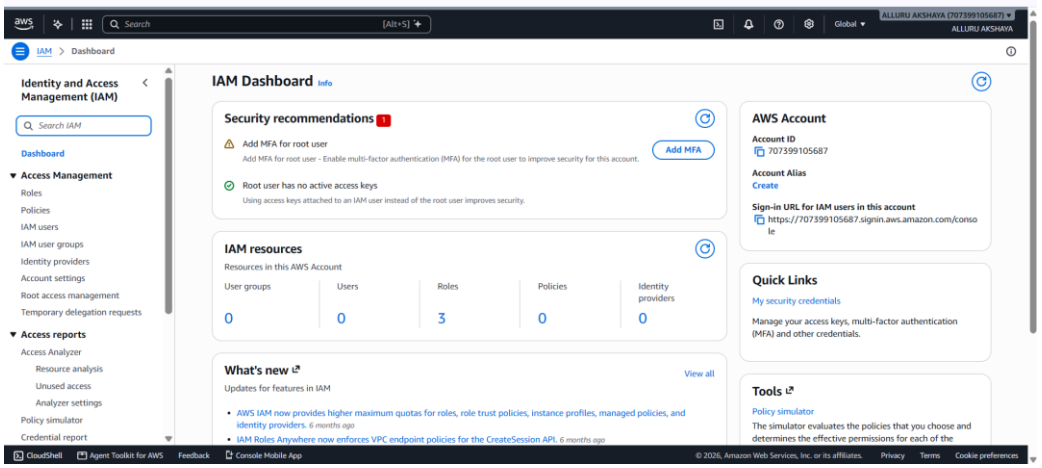
The objective of this task is to enable Multi-Factor Authentication (MFA) in an AWS account to improve account security and prevent unauthorized access. MFA provides an additional layer of protection by requiring users to verify their identity using two different authentication factors, such as a password and a one-time verification code generated by an authenticator application.

In AWS, MFA can be enabled for the root user and IAM users through the AWS Management Console. An authenticator application such as Google Authenticator or Microsoft Authenticator can be used to generate time-based one-time passwords (TOTP).

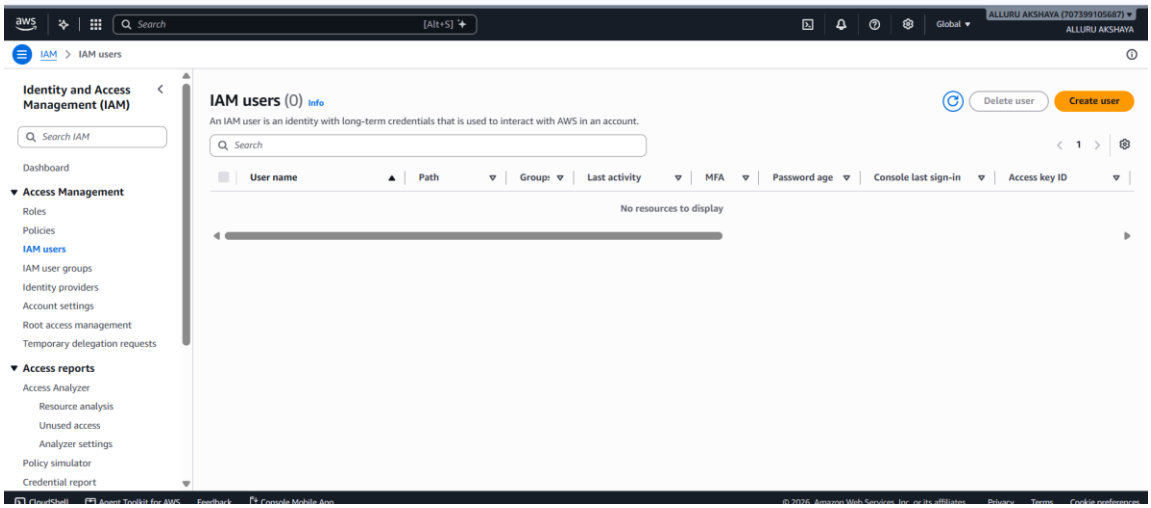
Steps:

Part A: Creating an IAM User

1. Open a web browser and navigate to the AWS Management Console.
2. Sign in to the AWS account using the root user credentials.
3. In the AWS Management Console search bar, type IAM and select Identity and Access Management (IAM).



4. From the left navigation pane, select Users.
5. Click Create user.

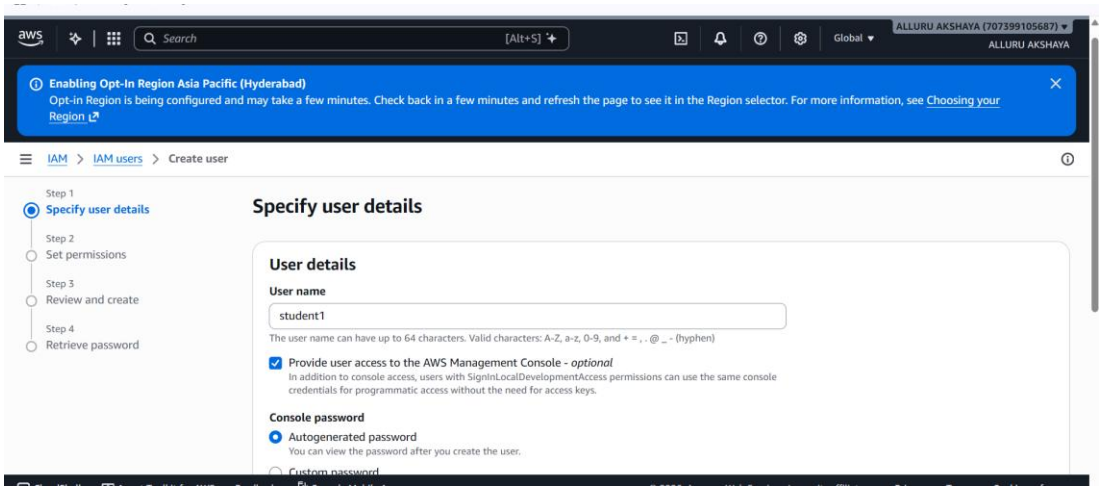


6. Enter a username (e.g., StudentUser01).

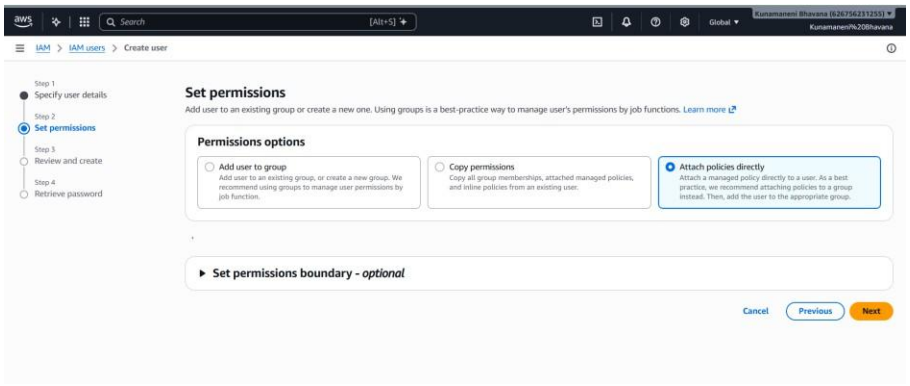
7. If the console provides an option to enable console access, select Provide user access to the AWS Management Console.

8. Set a custom password or allow AWS to generate a password (Autogenerated password.)

9. Select the option requiring the user to create a new password at the next sign-in, if appropriate. (If required you can opt). Click Next.



10. Under Set permissions, select Attach policies directly.



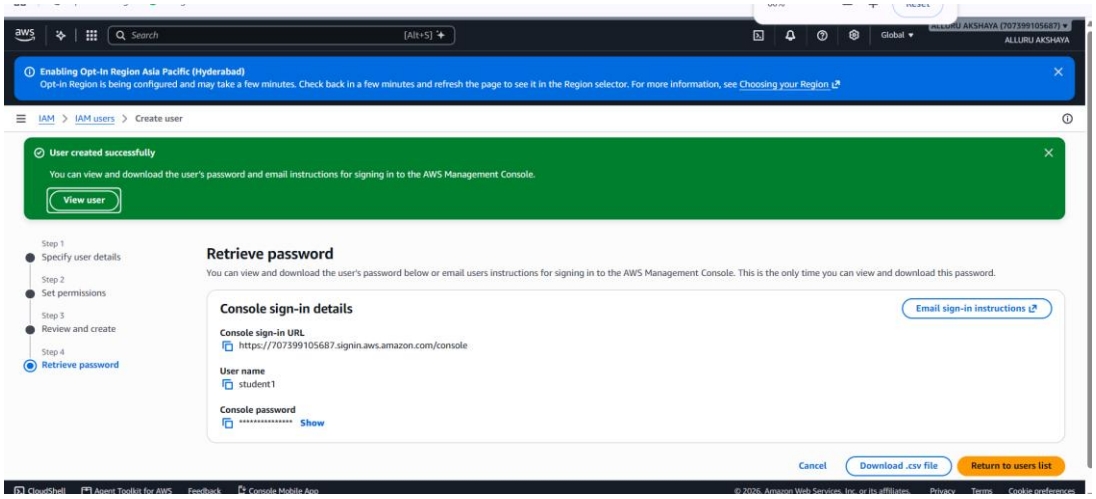
11. Select AdministratorAccess - AWS managed - job functionunder policy name and type. Click Next.

12. Review the username and permission details.

(Tags are optional. You can add if required.)

13. Click Create user.

14. Note down the IAM user sign-in details and securely save the generated password, if applicable. (You can download .csv file)



15. You can view to see the created IAM user when you click Return to user.

16. Sign out of the root account.

Part B: Signing in as an IAM User

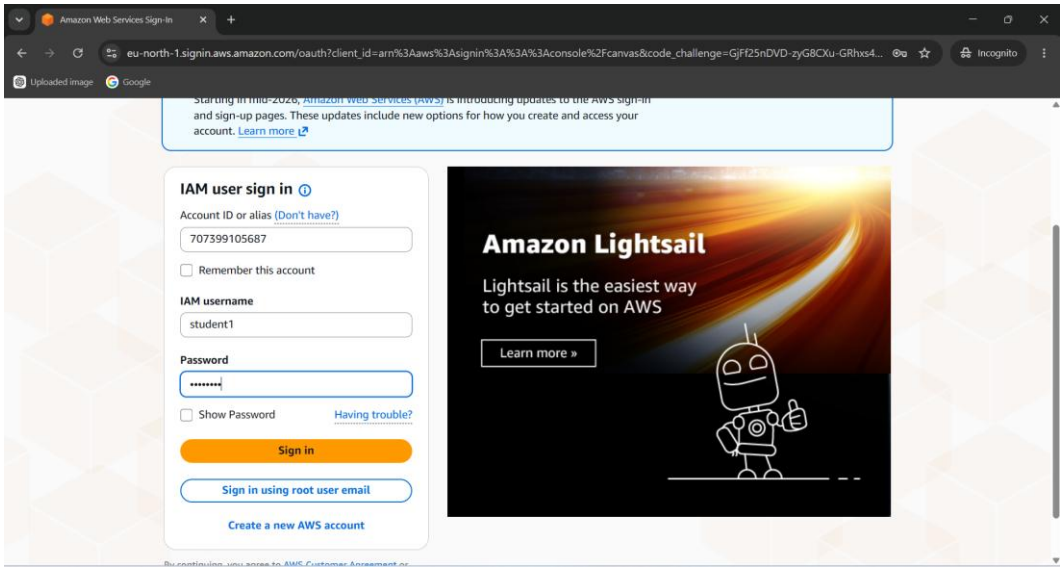
1. Open the AWS Management Console sign-in page

2. Select IAM user as the sign-in type.

Or instead of steps 1 and 2 .(Use the Console sign-in URL saved in the downloaded csv file)

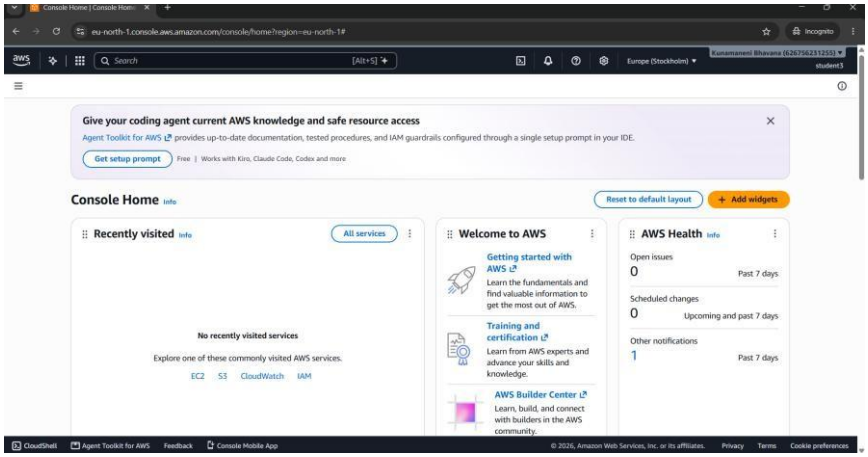
3. Enter the AWS account ID or account alias. (automatically seen if Console sign-in URL from downloaded csv file is used)

4. Enter the newly created IAM username and password. (present in the downloaded csv file)



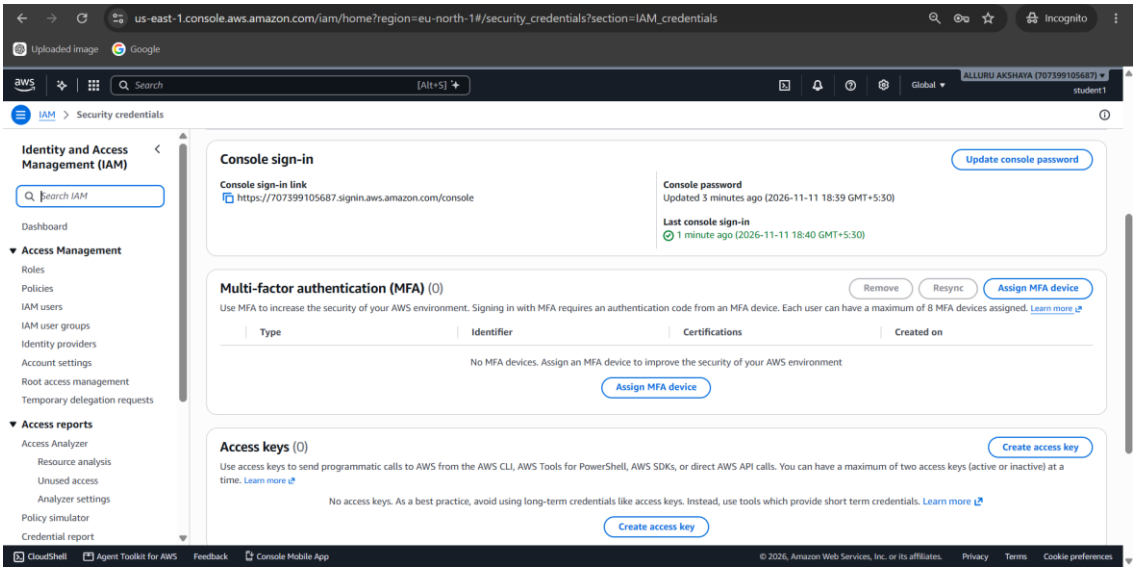
5. If prompted, change the temporary password and set a new password.

6. Verify that you can successfully access the AWS Management Console.

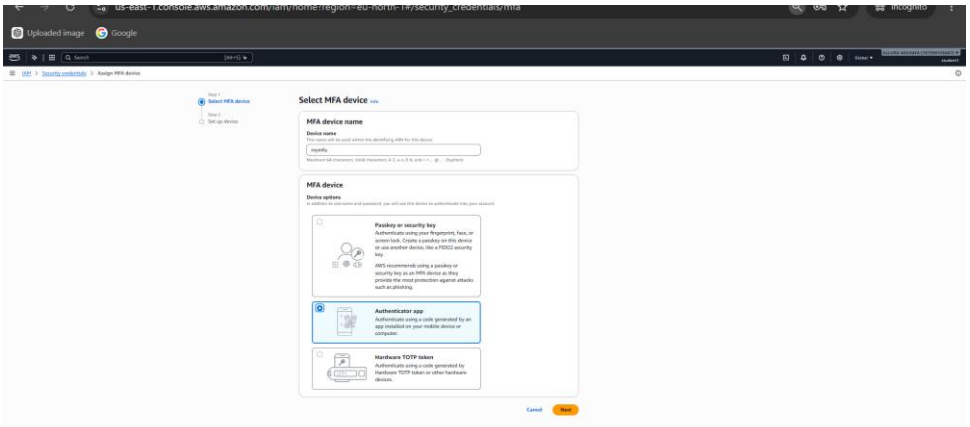


Part C: Enabling Multi-Factor Authentication (MFA)

1. Sign in to the AWS Management Console using the IAM user credentials.
2. Search for IAM in the AWS Management Console search bar.
3. Open the IAM dashboard.
4. From the left navigation pane, select Users.
5. Click the newly created IAM username.
6. Open the Security credentials tab.

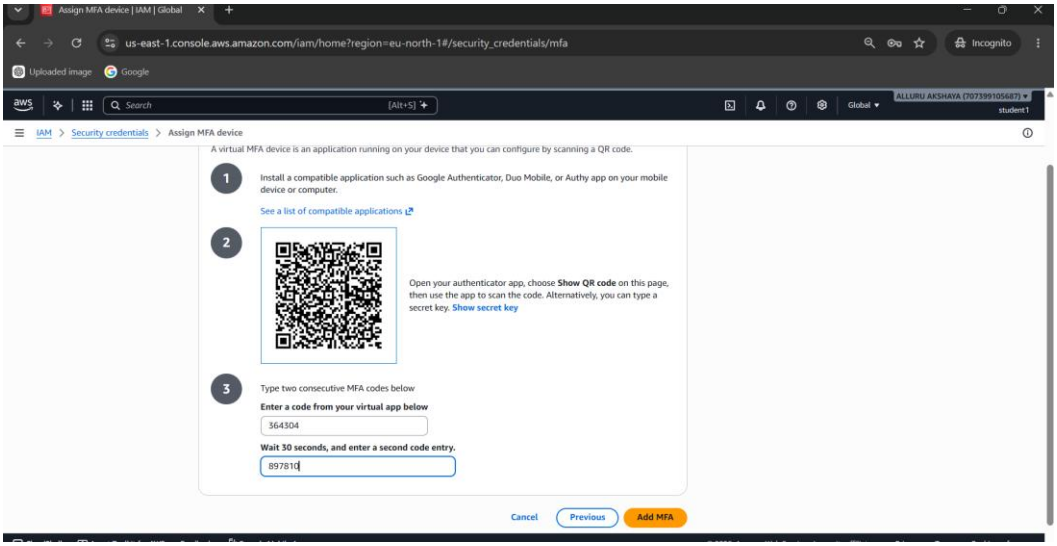


7. Locate the Multi-Factor Authentication (MFA) section.
8. Click Assign MFA device.
9. Enter a device name (e.g., StudentMFA01).
10. Select the MFA device type:
    - Passkey or Security key
    - Authenticator app (Google Authenticator or Microsoft Authenticator).

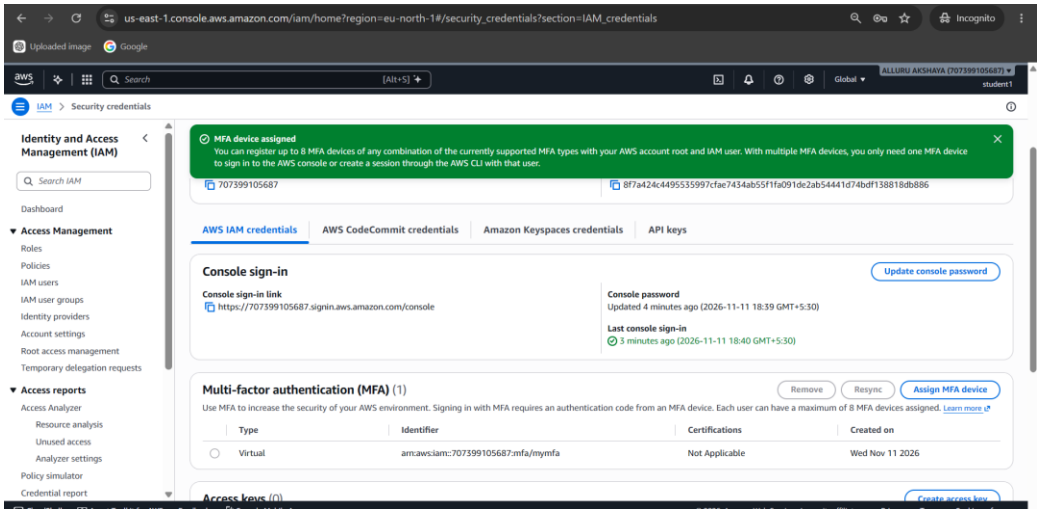


Option B: If you select Authenticator App

11. Select Authenticator app and click Next.
12. Open an authenticator application on your smartphone (Google Authenticator or Microsoft Authenticator).
13. Select the option to add a new account.
14. Scan the QR code displayed in the AWS console.
15. The authenticator application will generate a six-digit authentication code.
16. Enter the first six-digit MFA code displayed in the application.
17. Wait for the application to generate the next code.
18. Enter the second six-digit MFA code.



19. Click Add MFA or Assign MFA to complete the setup.
20. Verify that the selected MFA device is displayed as assigned to the IAM user.



Part D: Testing MFA Authentication

1. Sign out of the AWS Management Console.
2. Open the AWS sign-in page.
3. Select IAM user sign-in.
4. Enter the AWS account ID or account alias.
5. Enter the IAM username and password.
6. Complete the MFA verification based on the device configured:

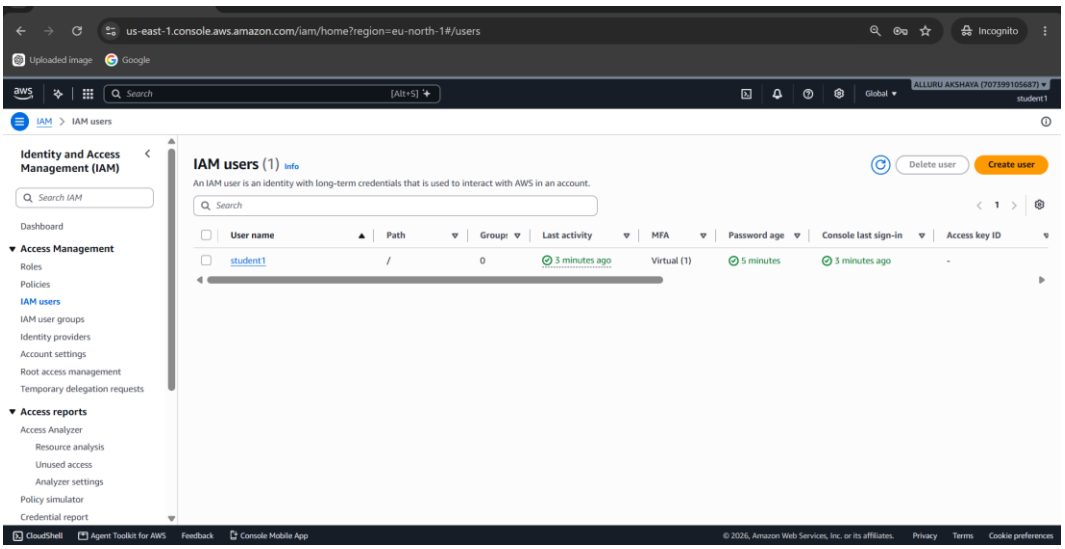
If Passkey or Security Key was selected:

- When prompted, select the option to sign in using the registered passkey.
- Complete the verification using the registered device, biometric authentication, or PIN.

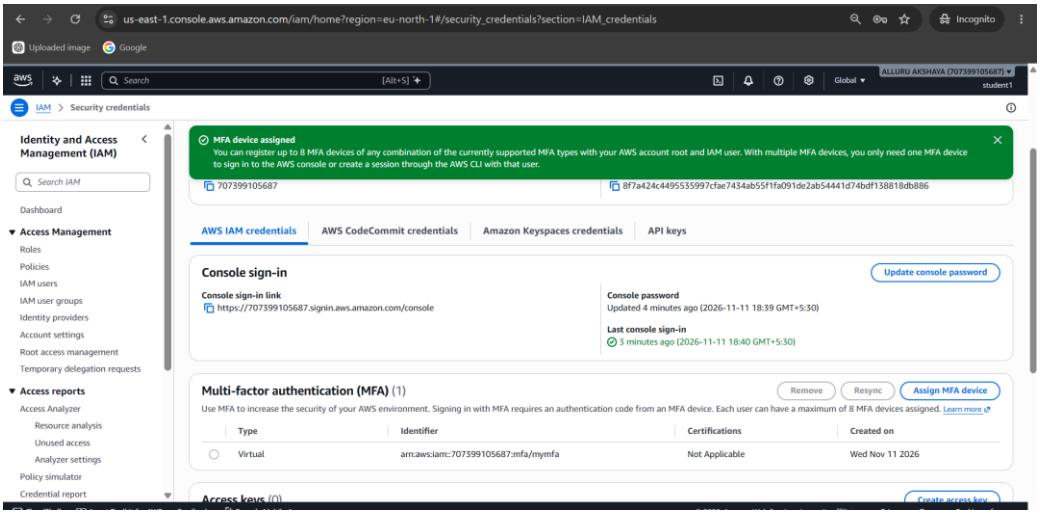
If Authenticator App was selected:

- When prompted, open the authenticator application.
- Enter the current six-digit MFA authentication code.

7. Verify that you can successfully log in to the AWS Management Console after completing MFA verification.



Output:


us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/security_credentials?section=IAM_credentials
Incognito
Uploaded image
Google
Search
[Alt+S]
Global
ALLURU AKSHAYA (707399105687)
student1
IAM > Security credentials
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
MFA device assigned
You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.
707399105687
8f7a424c4495535997cfae7434ab55f1fa091de2ab54441d74bdf138818db886
AWS IAM credentials
AWS CodeCommit credentials
Amazon Keyspaces credentials
API keys
Console sign-in
Update console password
Console sign-in link
https://707399105687.signin.aws.amazon.com/console
Console password
Updated 4 minutes ago (2026-11-11 18:39 GMT+5:30)
Last console sign-in
3 minutes ago (2026-11-11 18:40 GMT+5:30)
Multi-factor authentication (MFA) (1)
Remove
Resync
Assign MFA device
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
Virtual
arn:aws:iam::707399105687:mfa/mymfa
Not Applicable
Wed Nov 11 2026
Access keys (0)
Create access key